DATA SCIENCE & ARTIFICIAL INTELLIGENCE & CS/IT



Calculus and Optimization

Derivatives and Their Types

Lecture No. -03



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Recap of previous lecture



Topic

DERIVATIVES & THEIR TYPES

(PART-2)

Topics to be Covered



Topic DERIVATIVES & their Types

(Part 3)

- → Total Derivatives
- → Chain Rule of Partial Derivatives
- → JACOBIAN
- → Derivative of Homogeneous function.

RECAP

PARTIAL DIFF. [for $z = f(x,y)$]

$$\frac{\partial z}{\partial x} = \lim_{h \to 0} \left[\frac{f(x+h, y) - f(x,y)}{h} \right]_{y = \text{Const.}}$$

$$\frac{\partial z}{\partial y} = \lim_{k \to 0} \left[\frac{f(x, y+k) - f(x,y)}{k} \right]_{x = \text{Const.}}$$

Note:

In 2-D.

Equ^n of X axis : $y = 0$

Equ^n of line $\parallel^r$ to X axis : $y = k$

Equ^n of Y axis : $x = 0$

Equ^n of line $\parallel^r$ to Y axis : $x = h$

In 3D.

Equ^n of XY plane : $z = 0$

" " YZ plane : $x = 0$

" " ZX plane : $y = 0$

Equ^n of plane $\parallel^r$ to XZ plane : $y = k$

" " $\parallel^r$ to YZ plane : $x = h$

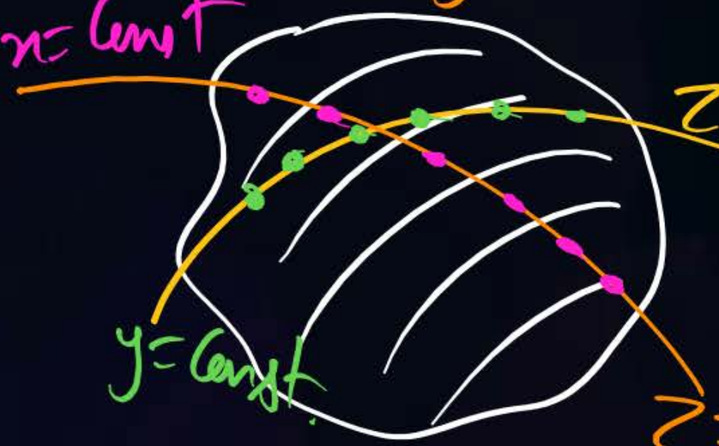
Significance of $\frac{\partial z}{\partial x}$ → $z=f(x,y)$

RECAP

If we cut our surface by the plane ⊥r to x z plane (i.e form $y=$ const) then we will get a curve of the type $z=f(x)$ & How

$\frac{\partial z}{\partial x}$ = Slope of tangent at any Random point on this curve.

Similarly we can define $\frac{\partial z}{\partial y}$ = ?



$z=f(x)$ (curve) ⇒ Slope of tangent on this curve.

$z=f(y)$ (curve) ⇒ Slope of tangent on this curve.

Note: $z = f(x,y)$ then $z_x = \frac{\partial z}{\partial x} = f_x = \frac{\partial f}{\partial x}$ all are same

(1) & $z_{xx} = f_{xx} = \frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 f}{\partial x^2}$

& $z_{xy} = f_{xy} = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = ?$

(2) $\boxed{\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}}$

(3) All the Results that are applicable in Case of ordinary Derivatives are also Valid in Case of Partial Derivatives keeping other Variable Constant

(4) Don't assume Dependent Variable as Constant, if we are solving Questions Based on Partial Derivatives.

(HW) if $r = x^2 + y - z$ & $y^3 + z^3 + yz - xy = 1$

where x & y are Independent Variables then at $(2, -1, 1)$ evaluate $\frac{\partial r}{\partial x} = ?$ (Ans = 4.5)

Sol:

Note:

(1) if $z = f(x, y)$ then $\frac{\partial z}{\partial x} = ?$ keeping y constant.

(2) if $z = f(x, y, u)$ then $\frac{\partial z}{\partial x} = ?$ $\because$ y & u as constant.

(3) if $w = f(x, y, z)$ then $\frac{\partial w}{\partial x} = ?$ $\because$ y & z as constant

(4) if we want to calculate $\frac{\partial^2 v}{\partial x \partial z} = ?$ then → Dep Variable = v, Ind Variables = x, z

Sol: $r = x^2 + y - z$ & $y^3 + z^3 + yz - xy = 1$

ie $r = f(x, y, z)$

$f(x, y, z) = c \Rightarrow z = f(x, y)$

$r \longrightarrow (x, y, z) \longrightarrow (x, y)$

$$\frac{\partial r}{\partial x} = \frac{\partial}{\partial x}(x^2 + y - z)$$

$$= 2x + 0 - \frac{\partial z}{\partial x}$$

$$\frac{\partial r}{\partial x} = 2x - \left[\frac{y}{3z^2 + y}\right]$$

At $(2, -1, 1)$, $\frac{\partial r}{\partial x} = 4 - \left(\frac{-1}{3-1}\right) = 4.5$

$$\frac{\partial}{\partial x}(y^3) + \frac{\partial}{\partial x}(z^3) + \frac{\partial}{\partial x}(yz) - \frac{\partial}{\partial x}(xy) = \frac{\partial}{\partial x}(1)$$

$$0 + \frac{\partial}{\partial z}(z^3)\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial x} - y(1) = 0$$

$$3z^2\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial x} = y$$

$$\frac{\partial z}{\partial x} = \frac{y}{3z^2 + y}$$

Homogeneous func^n :- Consider a func^n $\boxed{u = u(x, y)}$ then this func^n is called Homog. func^n of degree n, if each term is of same degree (n).

Trick: (i) $\boxed{u(\lambda x, \lambda y) = \lambda^n . u(x, y)}$ (ii) then u is called Homog func^n of degree 'n'. where $n \in R$.

eg $u = x^3 + 4xy^2 + 5x^2y$

$u(\lambda x, \lambda y) = (\lambda x)^3 + 4(\lambda x)(\lambda y)^2 + 5(\lambda x)^2(\lambda y)$

$= \lambda^3 (x^3 + 4xy^2 + 5x^2y)$

$= \lambda^3 \, u(x, y)$ Hence u is Homog func^n of degree 3

eg $u = \dfrac{x^2 + y^2}{\sqrt{x} + \sqrt{y}}$

$u(\lambda x, \lambda y) = \dfrac{(\lambda x)^2 + (\lambda y)^2}{\sqrt{\lambda x} + \sqrt{\lambda y}}$

$= \dfrac{\lambda^2}{\sqrt{\lambda}} \left[\dfrac{x^2 + y^2}{\sqrt{x} + \sqrt{y}} \right] = \lambda^{3/2} . u(x, y)$

ie u is Homog of degree $3/2$

eg $u = x^5 + 4x^2y^3 + \log(x^5)$

$= u(x, y)$

Now $u(\lambda x, \lambda y) = \lambda^5 x^5 + 4\lambda^2 x^2 \lambda^3 y^3 + 5\log(\lambda x)$

$= \lambda^5 \left[x^5 + 4x^2y^3 + \frac{5}{\lambda^5}\log(\lambda x) \right]$

$\neq \lambda^n u(x, y)$

i.e. u is Non Homog func^n of degree 5 ??

Blunder

eg $u = \log x - \log y$

$u = \log\left(\frac{x}{y}\right)$

Now $u(\lambda x, \lambda y) = \log\left(\frac{\lambda x}{\lambda y}\right)$

$= \lambda^0 \cdot \log\left(\frac{x}{y}\right)$

So u is Homog func^n of degree $n = 0$

Euler Theorem for Homog. func^n →

if u is Homog. func^n of degree n then $\boxed{x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=n.u}$

Note:-

① $x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=n(n-1)u$

② Sp. Note: if $\boxed{Z=u+v}$ where u & v are Homog. func^n of degree n_1 & n_2 But Z is Non Homog. func^n then.

(i) $x\frac{\partial Z}{\partial x}+y\frac{\partial Z}{\partial y}=\boxed{n_1u+n_2v}$

(ii) $x^2\frac{\partial^2 Z}{\partial x^2}+2xy\frac{\partial^2 Z}{\partial x\partial y}+y^2\frac{\partial^2 Z}{\partial y^2}=\boxed{n_1(n_1-1)u+n_2(n_2-1)v}$

Ex If $z = \dfrac{x^{\frac{1}{4}} + y^{\frac{1}{4}}}{x^{\frac{1}{3}} + y^{\frac{1}{3}}}$ then

① $x\dfrac{\partial z}{\partial x} + y\dfrac{\partial z}{\partial y} = ? = \dfrac{-1}{12}z = \dfrac{-z}{12}$

② $x^2 z_{xx} + 2xy z_{xy} + y^2 z_{yy} = ? = \dfrac{-1}{12}\left(\dfrac{-1}{12} - 1\right)z$

$= \dfrac{13}{144}z$

$z = z(x, y)$

Now $z(\lambda x, \lambda y) = \dfrac{\lambda^{\frac{1}{4}}}{\lambda^{\frac{1}{3}}}\left[\dfrac{x^{\frac{1}{4}} + y^{\frac{1}{4}}}{x^{\frac{1}{3}} + y^{\frac{1}{3}}}\right] = \lambda^{\frac{-1}{12}} \cdot z(x, y)$

ie z is Homog fnc^n of degree $n = \dfrac{-1}{12}$

Q. If $u = \log_e\left(\frac{x^2+y^2}{x+y}\right)$ then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = ?$

E. Th for u.

Sol: $\because u(\lambda x, \lambda y) \neq \lambda^n u(x,y)$ i.e. u is Non Homog func^n & we can't apply n

Let $e^u = \frac{x^2+y^2}{x+y} = v$ (let) i.e. $v = e^u = \frac{x^2+y^2}{x+y}$

$\because v(\lambda x, \lambda y) = \frac{\lambda^2}{\lambda}\left(\frac{x^2+y^2}{x+y}\right) = \lambda v(x,y)$ i.e. v is Homog func^n of $n=1$

So Applying E. Th for v i.e.

$$x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y} = n.v$$

$$x\frac{\partial}{\partial x}(e^u) + y\frac{\partial}{\partial y}(e^u) = 1.e^u$$

$$x\frac{\partial}{\partial u}(e^u)\frac{\partial u}{\partial x} + y\frac{\partial}{\partial u}(e^u)\frac{\partial u}{\partial y} = 1.e^u$$

$$x(e^u)\frac{\partial u}{\partial x} + y(e^u)\frac{\partial u}{\partial y} = e^u$$

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 1 \quad \text{Ans}$$

Q: if $u = \tan^{-1}\left(\frac{x^3+y^3}{x+y}\right)$ then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = ?$

Sol: $u(\lambda x, \lambda y) \neq \lambda^n u(x,y)$ so u is Hom Homog func^n.

Let $\boxed{V = \tan u} = \frac{x^3+y^3}{x+y}$; $\because V(\lambda x, \lambda y) = \lambda^2 V(x,y)$ is V is Homog $df^n = 2$

So for V: $x\frac{\partial V}{\partial x} + y\frac{\partial V}{\partial y} = nV$

$$x\frac{\partial}{\partial x}(\tan u) + y\frac{\partial}{\partial y}(\tan u) = 2\tan u$$

$$x(\sec^2 u)\frac{\partial u}{\partial x} + y(\sec^2 u)\frac{\partial u}{\partial y} = 2\tan u$$

is $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{2\tan u}{\sec^2 u}$

$= 2\frac{\sin u}{\cos u} \times \cos^2 u$

$= \sin 2u.$

Q: If $z = x^2y^4 \sin^{-1}\left(\frac{x}{y}\right) + \log x - \log y$ then $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = ?$

Sol: $z = x^2y^4 \sin^{-1}\left(\frac{x}{y}\right) + \log\left(\frac{x}{y}\right)$

let $z = u + v$ ——— (1)

where $u = x^2y^4 \sin^{-1}\left(\frac{x}{y}\right)$ is Homog func^n of degree $n_1 = 6$

i.e. $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 6u$ ——— (2)

Now $v = \log\left(\frac{x}{y}\right)$ is also Homog func^n of $n_2 = 0$

i.e. $x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y} = 0$ ——— (3)

Now $z = u + v$

$$\frac{\partial z}{\partial x} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial x}$$

$$\frac{\partial z}{\partial y} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}$$

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = x\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial x}\right) + y\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}\right)$$

$$= \left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right) + \left(x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y}\right)$$

$$= 6u + 0 = 6 \cdot x^2y^4 \sin^{-1}\left(\frac{x}{y}\right)$$

(ii) Also evaluate $x^2 z_{xx} + 2xy\, z_{xy} + y^2 z_{yy} = ?$

$$= n_1(n_1-1)u + n_2(n_2-1)v$$

$$= 6(6-1)u + 0(0-1)v$$

$$= 30u + 0$$

$$= 30x^2y^4 \sin^{-1}\left(\frac{x}{y}\right)$$ Ans

Derivative of an Implicit func^n

Consider $f(x,y)=C$ then $\boxed{\frac{dy}{dx}=-\frac{f_x}{f_y}}$

Proof: $df = d(C)$

$\left(\frac{\partial f}{\partial x}\right)dx+\left(\frac{\partial f}{\partial y}\right)dy=0 \quad (\because f \to x,y)$

$\Rightarrow \frac{dy}{dx}=-\frac{\partial f/\partial x}{\partial f/\partial y}$ Hence proved

(M-II) Consider $f \equiv x^3+y^3+3xy-1=0$

$\therefore \frac{dy}{dx}=-\frac{\partial f/\partial x}{\partial f/\partial y}=-\left[\frac{3x^2+3y}{3y^2+3x}\right]=\frac{-(x^2+y)}{y^2+x}$

Q: $\underbrace{x^3+y^3+3xy=1}_{f(x,y)=C}$ then $\frac{dy}{dx}=?$

(M-I)

$\frac{d}{dx}(x^3)+\frac{d}{dx}(y^3)+3\frac{d}{dx}(xy)=\frac{d}{dx}(1)$

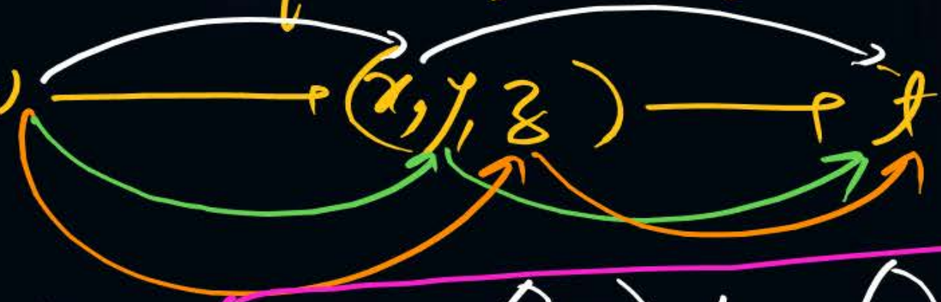
$3x^2+\frac{d}{dy}(y^3)\frac{dy}{dx}+3\left[x\frac{d(y)}{dx}+y(1)\right]=0$

$3x^2+3y^2\frac{dy}{dx}+3x\frac{dy}{dx}+3y=0$

$(3y^2+3x)\frac{dy}{dx}=-3x^2-3y$

$\frac{dy}{dx}=-\left(\frac{x^2+y}{y^2+x}\right)$

Total Derivative :→ if $w = f(x, y, z)$ where $x = x(t), y = y(t), z = z(t)$

i.e. 

Standard Result is: $$dw = \left(\frac{\partial w}{\partial x}\right) dx + \left(\frac{\partial w}{\partial y}\right) \cdot dy + \left(\frac{\partial w}{\partial z}\right) \cdot dz$$ → Learn by ❤

& Total Derivative of w with resp to t is given as.

$$\frac{dw}{dt} = \left(\frac{\partial w}{\partial x}\right) \frac{dx}{dt} + \left(\frac{\partial w}{\partial y}\right) \frac{dy}{dt} + \left(\frac{\partial w}{\partial z}\right) \frac{dz}{dt}$$

Note: if $w \longrightarrow (x, y, z) \longrightarrow x$ alone then $\frac{dw}{dx} = \left(\frac{\partial w}{\partial x}\right) + \left(\frac{\partial w}{\partial y}\right) \frac{dy}{dx} + \left(\frac{\partial w}{\partial z}\right) \frac{dz}{dx}$

Kalka Question:

Q: $d(xy) = ?$

(a) $dx\,dy$

(b) $dx + dy$

(c) $x\,dx + y\,dy$

(d) $x\,dy + y\,dx$

(M-I) $d(fg) = f g' + g f'$

$\Rightarrow d(xy) = x\,d(y) + y\,d(x)$

$= x\,dy + y\,dx$

(M-II) Let $f = xy$

i.e. $f \longrightarrow (x, y)$

So using Total Derivative Concept

$$df = \left(\frac{\partial f}{\partial x}\right)dx + \left(\frac{\partial f}{\partial y}\right)dy$$

$$d(xy) = (y)\,dx + (x)\,dy$$

Note:

$$\frac{d}{dx}(xy) = x\frac{dy}{dx} + y(1)$$

$$\frac{d}{dy}(xy) = x(1) + y\frac{dx}{dy}$$

$$d(xy) = x\,d(y) + y\,d(x)$$

$$= x\,dy + y\,dx$$

Q: If $u = x^2 - y^2 + 2\sin yz$ then

$\frac{\partial u}{\partial x} = ? = 2x - 0 + 0 = 2x$

$\frac{\partial u}{\partial y} = ? = 0 - 2y + 2\cos(yz)\frac{\partial}{\partial y}(yz)$

$= -2y + 2z\cos(yz)$

$\frac{\partial u}{\partial z} = ? = 0 - 0 + 2\cos(yz)\frac{\partial}{\partial z}(yz)$

$= 2y\cos(yz)$

Q: If $y = e^x$ then

$\frac{dy}{dx} = ? = e^x$

Q: If $z = \ln x$ then

$\frac{dz}{dx} = ? = \frac{1}{x}$

Q. if $u = x^2 - y^2 + 2\sin yz$ where $y = e^x$, $z = \ln x$ then $\frac{du}{dx} = ?$

Sol: $u \longrightarrow (x, y, z) \longrightarrow x$ alone. is 

Question is Based on Total Derivative Concept.

$$\boxed{du = \left(\frac{\partial u}{\partial x}\right)dx + \left(\frac{\partial u}{\partial y}\right)dy + \left(\frac{\partial u}{\partial z}\right)dz}$$ Standard Result.

$$\Rightarrow \frac{du}{dx} = \left(\frac{\partial u}{\partial x}\right) + \left(\frac{\partial u}{\partial y}\right)\frac{dy}{dx} + \left(\frac{\partial u}{\partial z}\right)\frac{dz}{dx}$$

$$= (2x) + (-2y + 2z\cos yz)\,e^x + (2y\cos yz)\cdot\frac{1}{x}$$

Q. If $u = x\log(xy)$ where $x^3+y^3+3xy=1$ & x is an Ind Variable then $\frac{du}{dx}=?$

$f(x,y)=c \Rightarrow y=f(x)$

Sol: $u \longrightarrow (x,y) \longrightarrow x$ alone.

(ie our Question is Based on Total Derivative.)

$$du=\left(\frac{\partial u}{\partial x}\right)dx+\left(\frac{\partial u}{\partial y}\right)dy$$

$$\Rightarrow \frac{du}{dx}=\left(\frac{\partial u}{\partial x}\right)+\left(\frac{\partial u}{\partial y}\right)\frac{dy}{dx}$$

$u=x\log xy$

$\frac{\partial u}{\partial y}=x\left(\frac{1}{xy}\frac{\partial}{\partial y}(xy)\right)=\frac{x}{y}$

$\frac{\partial u}{\partial x}=x\left(\frac{1}{xy}y\right)+\log(xy)(1)$

$=1+\log xy$

$$=(1+\log xy)+\left(\frac{x}{y}\right)\left\{-\left(\frac{x^2+y}{y^2+x}\right)\right\}$$ Ans

Q. if $u = x\log(xy)$ then Find

(1) $\frac{\partial u}{\partial x} = ? = 1 + \log xy$

(2) $\frac{\partial u}{\partial y} = ? = \frac{x}{y}$

(3) Total derivative of $u = ?$

$$du = \left(\frac{\partial u}{\partial x}\right)dx + \left(\frac{\partial u}{\partial y}\right)dy$$

$$\Rightarrow du = (1 + \log xy)\,dx + \left(\frac{x}{y}\right)dy$$

Type IV Chain Rule of Partial Derivative + (Change of Variable Concept) →

If $u = f(r, s, t)$ where $r = r(x, y, z)$, $s = s(x, y, z)$, $t = t(x, y, z)$

i.e. $u \longrightarrow (r, s, t) \longrightarrow (x, y, z)$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\left(\frac{\partial r}{\partial x}\right) + \frac{\partial u}{\partial s}\left(\frac{\partial s}{\partial x}\right) + \frac{\partial u}{\partial t}\left(\frac{\partial t}{\partial x}\right)$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\left(\frac{\partial r}{\partial y}\right) + \frac{\partial u}{\partial s}\left(\frac{\partial s}{\partial y}\right) + \frac{\partial u}{\partial t}\left(\frac{\partial t}{\partial y}\right)$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r}\left(\frac{\partial r}{\partial z}\right) + \frac{\partial u}{\partial s}\left(\frac{\partial s}{\partial z}\right) + \frac{\partial u}{\partial t}\left(\frac{\partial t}{\partial z}\right)$$

Shortcut to learn above Result :-

$$u \longrightarrow (r, s, t) \longrightarrow (x, y, z)$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial x} + \frac{\partial u}{\partial t}\frac{\partial t}{\partial x} = \frac{\partial u}{\partial r}\left(\frac{\partial r}{\partial x}\right) + \frac{\partial u}{\partial s}\left(\frac{\partial s}{\partial x}\right) + \frac{\partial u}{\partial t}\left(\frac{\partial t}{\partial x}\right)$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial y} + \frac{\partial u}{\partial t}\frac{\partial t}{\partial y} = \frac{\partial u}{\partial r}\left(\frac{\partial r}{\partial y}\right) + \frac{\partial u}{\partial s}\left(\frac{\partial s}{\partial y}\right) + \frac{\partial u}{\partial t}\left(\frac{\partial t}{\partial y}\right)$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial z} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial z} + \frac{\partial u}{\partial t}\frac{\partial t}{\partial z} = \frac{\partial u}{\partial r}\left(\frac{\partial r}{\partial z}\right) + \frac{\partial u}{\partial s}\left(\frac{\partial s}{\partial z}\right) + \frac{\partial u}{\partial t}\left(\frac{\partial t}{\partial z}\right)$$

Q: if $u = f(x,y)$ where $x = r+s$ & $y = r-s$ then $\frac{\partial u}{\partial r} + \frac{\partial u}{\partial s} = ?$

Sol: $u \longrightarrow (x,y) \longrightarrow (r,s)$ i.e Question is Based on Chain Rule Concept.

$$u_r = \frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r} = u_x(1) + u_y(1)$$

$$u_s = \frac{\partial u}{\partial s} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial s} = u_x(1) + u_y(-1)$$

$$u_r + u_s = 2u_x + 0 = 2\frac{\partial u}{\partial x}$$ Ans

(a) $2u_x$ (b) $2u_y$
(c) u_x (d) $2u_r$

HW If $u = f\left(\frac{y+x}{xy}, \frac{z+x}{zx}\right)$ then $x^2u_x + y^2u_y + z^2u_z = ?$

r $\quad$ s

Ans = 0

Sol: $u = f(r,s)$ where $r = \frac{1}{x}+\frac{1}{y}$ & $s = \frac{1}{x}+\frac{1}{z}$

ie $u \longrightarrow (r,s) \longrightarrow (x,y,z)$

$u_x = ?$

$u_y = ?$

$u_z = ?$

THANK - YOU

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